Research Analysis Report

Research Topic: Agentic AI

Total Papers: 5

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Executive Summary

This report provides a technical overview of 'Agentic AI' including methodologies, research gaps and experiment planning.

Literature Review

Introduction: This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

Paper Summaries

• {'title': 'Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey', 'year': 2025, 'summary': 'Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention. Unlike traditional AI, which depends on structured instructions and close oversight, Agentic AI demonstrates adaptability, advanced decision-making capabilities and self-sufficiency, enabling it to operate dynamically in evolving environments. This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI. We examine its current and potential applications across various fields, including healthcare, finance, and adaptive software systems, emphasizing the advantages of deploying agentic systems in real-world scenarios. The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability. We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts. This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively.'}

• {'title': 'AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges', 'year': 2025, 'summary': 'This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven and enabled by LLMs and LIMs for taskspecific automation. Generative AI is positioned as a precursor providing the foundation, with AI agents advancing through tool integration, prompt engineering, and reasoning enhancements. We then characterize Agentic AI systems, which, in contrast to AI Agents, represent a paradigm shift marked by multi-agent collaboration, dynamic task decomposition, persistent memory, and coordinated autonomy. Through a chronological evaluation of architectural evolution, operational mechanisms, interaction styles, and autonomy levels, we present a comparative analysis across both AI agents and agentic AI paradigms. Application domains enabled by AI Agents such as customer support, scheduling, and data summarization are then contrasted with Agentic AI deployments in research automation, robotic coordination, and medical decision support. We further examine unique challenges in each paradigm including hallucination, brittleness, emergent behavior, and coordination failure, and propose targeted solutions such as ReAct loops, retrieval-augmented generation (RAG), automation coordination layers, and causal modeling. This work aims'}

• {'title': 'Small Language Models are the Future of Agentic AI', 'year': 2025, 'summary': 'Large language models (LLMs) are often praised for exhibiting near-human performance on a wide range of tasks and valued for their ability to hold a general conversation. The rise of agentic AI systems is, however, ushering in a mass of applications in which language models perform a small number of specialized tasks repetitively and with little variation. Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI. Our argumentation is grounded in the current level of capabilities exhibited by SLMs, the common architectures of agentic systems, and the economy of LM deployment. We further argue that in situations where general-purpose conversational abilities are essential, heterogeneous agentic systems (i.e., agents invoking multiple different models) are the natural choice. We discuss the potential barriers for the adoption of SLMs in agentic systems and outline a general LLM-to-SLM agent conversion algorithm. Our position, formulated as a value statement, highlights the significance of the operational and economic impact even a partial shift from LLMs to SLMs is to have on the AI agent industry. We aim to stimulate the'}

• {'title': 'The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges', 'year': 2025, 'summary': 'Agentic AI systems are a recently emerged and important approach that goes beyond traditional AI, generative AI, and autonomous systems by focusing on autonomy, adaptability, and goal-driven reasoning. This study provides a clear review of agentic AI systems by bringing together their definitions, frameworks, and architectures, and by comparing them with related areas like generative AI, autonomic computing, and multi-agent systems. To do this, we reviewed 143 primary studies on current LLM-based and non-LLM-driven agentic systems and examined how they support planning, memory, reflection, and goal pursuit. Furthermore, we classified architectural models, input–output mechanisms, and applications based on their task domains where agentic AI is applied, supported using tabular summaries that highlight real-world case studies. Evaluation metrics were classified as qualitative and quantitative measures, along with available testing methods of agentic AI systems to check the system’s performance and reliability. This study also highlights the main challenges and limitations of agentic AI, covering technical, architectural, coordination, ethical, and security issues. We organized the conceptual foundations, available tools, architectures, and evaluation metrics in this research, which defines a structured foundation for understanding and advancing agentic AI. These findings aim to help researchers and developers build better, clearer, and more adaptable systems'}

• {'title': 'TRiSM for Agentic AI: A Review of Trust, Risk, and Security Management in LLM-based Agentic Multi-Agent Systems', 'year': 2025, 'summary': 'Agentic AI systems, built upon large language models (LLMs) and deployed in multi-agent configurations, are redefining intelligence, autonomy, collaboration, and decision-making across enterprise and societal domains. This review presents a structured analysis of Trust, Risk, and Security Management (TRiSM) in the context of LLM-based Agentic Multi-Agent Systems (AMAS). We begin by examining the conceptual foundations of Agentic AI and highlight its architectural distinctions from traditional AI agents. We then adapt and extend the AI TRiSM framework for Agentic AI, structured around key pillars: \\textit{ Explainability, ModelOps, Security, Privacy} and \\textit{their Lifecycle Governance}, each contextualized to the challenges of AMAS. A risk taxonomy is proposed to capture the unique threats and vulnerabilities of Agentic AI, ranging from coordination failures to prompt-based adversarial manipulation. To support practical assessment in Agentic AI works, we introduce two novel metrics: the Component Synergy Score (CSS), which quantifies the quality of inter-agent collaboration, and the Tool Utilization Efficacy (TUE), which evaluates the efficiency of tool use within agent workflows. We further discuss strategies for improving explainability in Agentic AI, as well as approaches to enhancing security and privacy through encryption, adversarial robustness, and regulatory compliance. The review concludes with a research roadmap for the responsible development'}

Research Trends

- adaptability
- adversarial
- agent
- agentic
- amas
- application
- architectural
- architecture
- argumentation
- artificial
- automation
- autonomous
- autonomy
- beneficial
- capability
- challenge
- characteristic
- classified
- collaboration
- consideration
- constraint
- conversation
- coordination
- decision-making
- enabled
- ethical
- evaluation
- explainability
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- inter-agent
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- llm-based
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- multi-agent
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- security
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- slms
- structured
- support
- survey
- tasks
- then

Research Gaps

- We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts.

Future Scope

- We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts.

Conclusion: Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

Methodology Comparison

Total Papers: 5

Comparison

• {'title': 'Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey', 'research_area': 'Agentic AI', 'research_problem': 'The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability.', 'methodology': ['Survey', 'Framework'], 'keywords': ['agentic', 'autonomous', 'decision-making', 'real-world', 'self-sufficiency', 'adaptability', 'ethical', 'survey', 'application', 'artificial', 'beneficial', 'capability', 'characteristic', 'consideration', 'constraint'], 'year':

2025, 'citation_count': 693, 'quality_score': 100, 'quality_classification': 'Excellent'}

• {'title': 'AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges', 'research_area': 'Retrieval-Augmented Generation', 'research_problem': 'This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities.', 'methodology': ['Architecture', 'Experimental Evaluation', 'Large Language Model', 'Retrieval-Augmented Generation'], 'keywords': ['agent', 'multi-agent', 'memory', 'reasoning', 'retrieval-augmented', 'agentic', 'automation', 'coordination', 'paradigm', 'application', 'autonomy', 'challenge', 'enabled', 'support', 'then'], 'year': 2025, 'citation_count': 518, 'quality_score': 100, 'quality_classification': 'Excellent'}

• {'title': 'Small Language Models are the Future of Agentic AI', 'research_area': 'Agentic AI', 'research_problem': 'The rise of agentic AI systems is, however, ushering in a mass of applications in which language models perform a small number of specialized tasks repetitively and with little variation.', 'methodology': ['Architecture', 'Algorithm', 'Large Language Model'], 'keywords': ['agent', 'agentic', 'general-purpose', 'llm-to-slm', 'near-human', 'slms', 'language', 'llms', 'position', 'tasks', 'application', 'architecture', 'argumentation', 'capability', 'conversation'], 'year': 2025, 'citation_count': 354, 'quality_score': 90, 'quality_classification': 'Excellent'}

• {'title': 'The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges', 'research_area': 'Multi-Agent Systems', 'research_problem': 'This study also highlights the main challenges and limitations of agentic AI, covering technical, architectural, coordination, ethical, and security issues.', 'methodology': ['Framework', 'Architecture', 'Experimental Evaluation', 'Large Language Model'], 'keywords': ['multi-agent', 'agentic', 'autonomous', 'memory', 'planning', 'reasoning', 'goal-driven', 'non-llm-driven', 'real-world', 'llm-based', 'architectural', 'architecture', 'classified', 'evaluation', 'foundation'], 'year': 2025, 'citation_count': 170, 'quality_score': 90, 'quality_classification': 'Excellent'}

• {'title': 'TRiSM for Agentic AI: A Review of Trust, Risk, and Security Management in LLM-based Agentic Multi-Agent Systems', 'research_area': 'Multi-Agent Systems', 'research_problem': 'We then adapt and extend the AI TRiSM framework for Agentic AI, structured around key pillars: \\textit{ Explainability, ModelOps, Security, Privacy} and \\textit{their Lifecycle Governance}, each contextualized to the challenges of AMAS.', 'methodology': ['Framework', 'Architecture', 'Large Language Model'], 'keywords': ['multi-agent', 'agentic', 'agent', 'decision-making', 'inter-agent', 'prompt-based', 'llm-based', 'security', 'adversarial', 'collaboration', 'explainability', 'structured', 'amas', 'privacy', 'review'], 'year': 2025, 'citation_count': 106, 'quality_score': 90, 'quality_classification': 'Excellent'}

Highest Cited Paper

title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

authors: ['D. Acharya', 'Karthigeyan Kuppan', 'Divya Bhaskaracharya']

year: 2025

citation_count: 693

summary: Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention. Unlike traditional AI, which depends on structured instructions and close oversight, Agentic AI demonstrates adaptability, advanced decision-making capabilities and self-sufficiency, enabling it to operate dynamically in evolving environments. This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI. We examine its current and potential applications across various fields, including healthcare, finance, and adaptive software systems, emphasizing the advantages of deploying agentic systems in real-world scenarios. The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability. We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts. This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively.

research_problem: The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability.

methodology: ['Survey', 'Framework']

key_contributions: ['Unlike traditional AI, which depends on structured instructions and close oversight, Agentic AI demonstrates adaptability, advanced decision-making capabilities and self-sufficiency, enabling it to operate dynamically in evolving environments.']

future_work: ['We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts.']

keywords: ['agentic', 'autonomous', 'decision-making', 'real-world', 'self-sufficiency', 'adaptability', 'ethical', 'survey', 'application', 'artificial', 'beneficial', 'capability', 'characteristic', 'consideration', 'constraint']

research_area: Agentic AI

paper_score: 100

paper_quality: Excellent

Latest Paper

title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

authors: ['D. Acharya', 'Karthigeyan Kuppan', 'Divya Bhaskaracharya']

year: 2025

citation_count: 693

summary: Agentic AI, an emerging paradigm in artificial intelligence, refers to autonomous systems designed to pursue complex goals with minimal human intervention. Unlike traditional AI, which depends on structured instructions and close oversight, Agentic AI demonstrates adaptability, advanced decision-making capabilities and self-sufficiency, enabling it to operate dynamically in evolving environments. This survey thoroughly explores the foundational concepts, unique characteristics, and core methodologies driving the development of Agentic AI. We examine its current and potential applications across various fields, including healthcare, finance, and adaptive software systems, emphasizing the advantages of deploying agentic systems in real-world scenarios. The paper also addresses the ethical challenges posed by Agentic AI, proposing solutions for goal alignment, resource constraints, and environmental adaptability. We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts. This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively.

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methodology: ['Survey', 'Framework']

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future_work: ['We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts.']

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research_area: Agentic AI

paper_score: 100

paper_quality: Excellent

Common Methodologies

• Algorithm

• Architecture

• Experimental Evaluation

• Framework

• Large Language Model

• Retrieval-Augmented Generation

• Survey

Research Areas

• Agentic AI

• Multi-Agent Systems

• Retrieval-Augmented Generation

Common Keywords

• adaptability

• adversarial

• agent

• agentic

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• then

Methodology Frequency

Architecture: 4

Large Language Model: 4

Framework: 3

Experimental Evaluation: 2

Survey: 1

Retrieval-Augmented Generation: 1

Algorithm: 1

Most Common Methodology: Architecture

Quality Distribution

Excellent: 5

Publication Years

2025: 5

Citation Statistics

highest: 693

lowest: 106

average: 368.2

total: 1841

Comparison Summary

total_papers: 5

most_common_methodology: Architecture

research_areas: ['Agentic AI', 'Multi-Agent Systems', 'Retrieval-Augmented Generation']

quality_distribution: {'Excellent': 5}

citation_statistics: {'highest': 693, 'lowest': 106, 'average': 368.2, 'total': 1841}

highest_cited_title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

latest_paper_title: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

Research Gap Analysis

Total Papers: 5

Research Areas

• Agentic AI

- Multi-Agent Systems
- Retrieval-Augmented Generation

Common Keywords

- agentic
- agent
- application
- multi-agent
- architecture
- autonomous
- capability
- decision-making
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- real-world
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Future Work

- We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts.

Research Area Distribution

Agentic AI: 2

Multi-Agent Systems: 2

Retrieval-Augmented Generation: 1

Keyword Frequency

agentic: 5

agent: 3

application: 3

multi-agent: 3

architecture: 2

autonomous: 2
capability: 2
decision-making: 2
llm-based: 2
memory: 2
real-world: 2
reasoning: 2
adaptability: 1
adversarial: 1
amas: 1
architectural: 1
argumentation: 1
artificial: 1
automation: 1
autonomy: 1
beneficial: 1
challenge: 1
characteristic: 1
classified: 1
collaboration: 1
consideration: 1
constraint: 1
conversation: 1
coordination: 1
enabled: 1
ethical: 1
evaluation: 1
explainability: 1
foundation: 1
general-purpose: 1
goal-driven: 1
inter-agent: 1
language: 1
llm-to-slm: 1
llms: 1
near-human: 1
non-llm-driven: 1
paradigm: 1

planning: 1

position: 1

privacy: 1

prompt-based: 1

retrieval-augmented: 1

review: 1

security: 1

self-sufficiency: 1

slms: 1

structured: 1

support: 1

survey: 1

tasks: 1

then: 1

Research Trends

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- multi-agent
- architecture
- autonomous
- capability
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- reasoning
- adaptability
- adversarial
- amas

Gap Categories

datasets: []

models: ['This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven and enabled by LLMs and LIMs for taskspecific automation. Generative AI is positioned as a precursor providing the foundation, with AI agents advancing through tool integration, prompt engineering, and reasoning enhancements. We then characterize Agentic AI systems, which, in contrast to AI Agents, represent a paradigm shift marked by multi-agent collaboration, dynamic task decomposition, persistent memory, and coordinated autonomy. 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The rise of agentic AI systems is, however, ushering in a mass of applications in which language models perform a small number of specialized tasks repetitively and with little variation. Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI. Our argumentation is grounded in the current level of capabilities exhibited by SLMs, the common architectures of agentic systems, and the economy of LM deployment. We further argue that in situations where general-purpose conversational abilities are essential, heterogeneous agentic systems (i.e., agents invoking multiple different models) are the natural choice. We discuss the potential barriers for the adoption of SLMs in agentic systems and outline a general LLM-to-SLM agent conversion algorithm. 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knowledge_freshness: []

evaluation: ['This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven and enabled by LLMs and LIMs for taskspecific automation. Generative AI is positioned as a precursor providing the foundation, with AI agents advancing through tool integration, prompt engineering, and reasoning enhancements. We then characterize Agentic AI systems, which, in contrast to AI Agents, represent a paradigm shift marked by multi-agent collaboration, dynamic task decomposition, persistent memory, and coordinated autonomy. 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We outline a framework for safely and effectively integrating Agentic AI into society, highlighting the need for further research on ethical considerations to ensure beneficial societal impacts. This survey serves as a comprehensive introduction to Agentic AI, guiding researchers, developers, and policymakers in engaging with its transformative potential responsibly and creatively.', 'AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges', 'This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven and enabled by LLMs and LIMs for taskspecific automation. 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Here we lay out the position that small language models (SLMs) are sufficiently powerful, inherently more suitable, and necessarily more economical for many invocations in agentic systems, and are therefore the future of agentic AI. Our argumentation is grounded in the current level of capabilities exhibited by SLMs, the common architectures of agentic systems, and the economy of LM deployment. We further argue that in situations where general-purpose conversational abilities are essential, heterogeneous agentic systems (i.e., agents invoking multiple different models) are the natural choice. We discuss the potential barriers for the adoption of SLMs in agentic systems and outline a general LLM-to-SLM agent conversion algorithm. Our position, formulated as a value statement, highlights the significance of the operational and economic impact even a partial shift from LLMs to SLMs is to have on the AI agent industry. 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Research Gaps

• Retrieval quality and the ability to consistently retrieve relevant information for complex queries remain important research challenges.

• Improved grounding, factuality, and reliable source attribution are required to reduce unsupported or incorrect generated information.

• Hallucination and factual reliability remain unresolved challenges in current AI systems.

• More robust and standardized evaluation and benchmarking methods are required.

• Current models and architectures still require improvements in reliability, adaptability, and generalization.

• Reliable coordination and communication between multiple agents remains an open research challenge.

• Security, privacy, trust, and robustness remain important unresolved challenges.

• Scalability, computational cost, and efficient resource usage remain challenges for practical deployment.

• Improved explainability and interpretability are required for reliable adoption of AI-based systems.

• Current systems require stronger reasoning capabilities for complex multi-step tasks.

Emerging Topics

• agentic

• agent

• application

• multi-agent

• architecture

• autonomous

• capability

• decision-making

• llm-based

• memory

• real-world

• reasoning

Recommendations

• Explore more reliable autonomous and multi-agent systems with improved coordination and task execution.

• Investigate improved inter-agent communication and coordination mechanisms for reliable multi-agent task execution.

• Improve reasoning capabilities through structured planning, multi-step inference, and verification mechanisms.

• Investigate more reliable autonomous planning and task decomposition techniques.

• Investigate long-term memory mechanisms for maintaining context across complex research and reasoning tasks.

• Improve retrieval quality and document selection for complex research queries.

• Strengthen source grounding and citation verification to improve factual reliability.

• Develop stronger hallucination detection and factual verification mechanisms.

• Evaluate proposed approaches using standardized benchmarks and reproducible evaluation protocols.

• Compare stronger model architectures and established baselines to identify improvements in reliability and generalization.

Recommendation Validation

• {'recommendation': 'Explore more reliable autonomous and multi-agent systems with improved coordination and task execution.', 'validation_status': 'Strongly Supported', 'evidence_score': 3, 'evidence': {'research_gaps': True, 'future_work': True, 'emerging_topics': True, 'methodology': False}}

• {'recommendation': 'Investigate improved inter-agent communication and coordination mechanisms for reliable multi-agent task execution.', 'validation_status': 'Strongly Supported', 'evidence_score': 3, 'evidence': {'research_gaps': True, 'future_work': True, 'emerging_topics': True,

'methodology': False}}

• {'recommendation': 'Improve reasoning capabilities through structured planning, multi-step inference, and verification mechanisms.', 'validation_status': 'Supported', 'evidence_score': 2, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': True, 'methodology': False}}

• {'recommendation': 'Investigate more reliable autonomous planning and task decomposition techniques.', 'validation_status': 'Weakly Supported', 'evidence_score': 0, 'evidence': {'research_gaps': False, 'future_work': False, 'emerging_topics': False, 'methodology': False}}

• {'recommendation': 'Investigate long-term memory mechanisms for maintaining context across complex research and reasoning tasks.', 'validation_status': 'Strongly Supported', 'evidence_score': 3, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': True, 'methodology': True}}

• {'recommendation': 'Improve retrieval quality and document selection for complex research queries.', 'validation_status': 'Supported', 'evidence_score': 2, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': True}}

• {'recommendation': 'Strengthen source grounding and citation verification to improve factual reliability.', 'validation_status': 'Partially Supported', 'evidence_score': 1, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': False}}

• {'recommendation': 'Develop stronger hallucination detection and factual verification mechanisms.', 'validation_status': 'Supported', 'evidence_score': 2, 'evidence': {'research_gaps': True, 'future_work': False, 'emerging_topics': False, 'methodology': True}}

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Summary

dominant_area: Agentic AI

top_trend: agentic

future_work_items: 1

research_gap_count: 10

validated_recommendation_count: 10

Experiment Plan

Research Areas

• Agentic AI

• Multi-Agent Systems

• Retrieval-Augmented Generation

Recommended Datasets

- AgentBench
- BEIR
- GAIA Benchmark
- HotpotQA
- Hugging Face Datasets
- MS MARCO
- Multi-Agent Benchmark
- Natural Questions
- OpenAI Evals
- TriviaQA

Baseline Models

- AutoGen
- BM25 Retrieval
- CrewAI
- Dense Retrieval RAG
- Hybrid Search RAG
- LangGraph
- Naive RAG
- ReAct
- Single-Agent LLM
- Vanilla RAG

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Retrieval Precision
- Retrieval Recall
- Context Relevance
- Answer Relevance
- Faithfulness
- Groundedness
- Task Success Rate
- Planning Accuracy
- Tool-Use Accuracy
- Agent Coordination Score
- ROC-AUC

Hardware Requirements

cpu: Intel Core i7 / AMD Ryzen 7

ram: 32 GB

gpu: NVIDIA RTX 3060 / RTX 4060 or better

storage: 200 GB SSD

Validation Strategy

• Train-Test Split
• K-Fold Cross Validation
• Hyperparameter Tuning
• Statistical Significance Testing
• Retrieval Evaluation
• Answer Quality Evaluation
• Task Success Evaluation
• Agent Coordination Evaluation
• Tool-Use Evaluation

Experimental Workflow

• Collect Knowledge Documents
• Preprocess and Chunk Documents
• Build Document Index
• Configure Retrieval System
• Retrieve Relevant Context
• Generate Grounded Response
• Evaluate Retrieval Quality
• Evaluate Answer Quality
• Compare Baseline and Proposed RAG
• Analyze Errors
• Draw Conclusions

Gap Based Recommendations

• Compare BM25, dense retrieval, and hybrid retrieval methods.
• Evaluate source attribution, faithfulness, and groundedness.
• Measure hallucination frequency and introduce factuality verification.
• Use standardized benchmarks and multiple evaluation metrics.
• Compare centralized and decentralized multi-agent coordination strategies.
• Evaluate robustness against prompt injection, adversarial inputs, and unauthorized tool usage.
• Measure computational cost, latency, resource consumption, and scalability.

• Evaluate explanation quality, transparency, and interpretability.

• Evaluate multi-step reasoning using complex benchmark tasks.

• Evaluate long-term memory and persistent context management.

• Evaluate long-horizon planning and decision-making performance.

• Validate the system in realistic real-world deployment scenarios.

Report Summary

Dominant Research Area: Agentic AI

Top Research Trend: agentic

Highest Cited Paper: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

Latest Paper: Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey

Recommended Datasets

• AgentBench

• BEIR

• GAIA Benchmark

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- Retrieval Recall
- Context Relevance
- Answer Relevance
- Faithfulness
- Groundedness
- Task Success Rate
- Planning Accuracy
- Tool-Use Accuracy
- Agent Coordination Score
- ROC-AUC

Conclusion

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.